

EE 630: HVDC Transmission and Flexible AC Transmission Systems

Course Code	EE 630
Course Name	HVDC Transmission and Flexible AC Transmission Systems
L-T-P-C	3-0-0-3
Prerequisites	EE-303 (Power systems) and EE-309 (Power electronics)
Intended for	PG
Distribution	Elective for M.S., M. Tech. & Ph.D.
Approval	recommended by the 37th BoA (Oct 2020)

Course Contents

- **Introduction to HVDC:** Introduction of DC power transmission technology, comparison of AC and DC transmission, limitation of HVDC transmission, reliability of HVDC systems, application of DC transmission, description of DC transmission system, planning for HVDC transmission, modern trends in DC transmission, advantages of HVDC. [5 Lectures]
- **Analysis of HVDC converters:** Simple rectifier circuits, required features of rectification circuits for HVDC transmission, choice of converter configuration, converter bridge characteristics. [4 Lectures]
- **Control of HVDC converter and systems:** Necessity of control of a DC link, rectifier control, compounding of rectifiers, power reversal of DC link, voltage dependent current order limit (VDCOL) characteristics of the converter, inverter extinction angle control, pulse phase control, starting and stopping of DC link, constant power control, control scheme of HVDC converters. [5 Lectures]
- **Harmonics and filters:** Generation of harmonics by converters, characteristics of harmonics on DC side, characteristics of current harmonics, characteristic variation of harmonic currents with variation of firing angle and overlap angle, effect of control mode on harmonics, non characteristic harmonic. [5 Lectures]
- **FACTS devices:** Introduction to FACTS; Compensation of transmission systems, Series and Shunt FACTS controllers - variable impedance type and switched converter type, Unified Power Flow Controller and other types of FACTS devices. [5 Lectures]
- **Load flow and stability analysis:** Component Models for the Analysis of AC DC Systems, Power flow analysis of AC-DC systems, Transient stability analysis, Dynamic stability analysis. Application of FACTS controllers in improvement of power system operation and stability. [5 Lectures]
- **Reactive power control:** Reactive power requirements in steady state, sources of reactive power, static VAR systems, reactive power control during transients. [3 Lectures]
- **Fault and protection schemes in HVDC systems:** Nature and types of faults, faults on AC side of the converter stations, converter faults, fault on DC side of the systems, protection against over currents and over voltages, protection of filter units. [5 Lectures]
- **Multiterminal HVDC systems:** Types of multiterminal (MTDC) systems, parallel operation aspect of MTDC. Control of power in MTDC. Power upgrading and conversion of AC lines into DC lines, Parallel AC/DC systems, FACTS and HVDC system application in wind power generation- VSC based applications for wind power systems. [5 Lectures]

Text Books

1. K. R. Padiyar, **HVDC Power Transmission Systems**, Wiley, 1990
2. J. Arrillaga, **High Voltage Direct Current Transmission**, The Institution of Electrical Engineers, 1998.
3. N. G. Hingorani, **Understanding FACTS**, IEEE Press, 2001.

References

1. K. R. Padiyar, **FACTS Controllers in Power Transmission and Distribution**, New age international, 2007.
2. E. W. Kimbark, **Direct Current Transmission**, Wiley-Interscience, 1971.
3. S N Singh, **Electric Power Generation, Transmission and Distribution**, 2nd Edition, PHI, 2008.
4. T J E Miller, **Reactive Power Control in Electric Systems**, Wiley, 1982.